BFW STEAM HEATER VER 1

Operator Training Simulator · E-101

Chulalongkorn University · Department of Chemical Engineering

RELIEF LOAD BASIS AND RELIEF DEVICE SIZING

DOCUMENT NO. : BFWH-V1-PE-RLB-001

DISCIPLINE : Process (relief)

PROJECT : BFW Steam Heater Ver 1 · E-101 Operator Training Simulator

STATUS : Issued for teaching (IFT)

API 521 screening of the overpressure and vacuum cases of the E-101 shell and tube side, relief loads, API 520 sizing and the relief device register. Tube rupture governs (22,452 kg/h BFW liquid); PSV-100 is sized for external fire (557 kg/h steam, orifice G); a rupture disc RD-100 is recommended for the tube rupture (HOLD).

Rev	Date	Description	Originator	Checker	Approver
0	25-Oct-26	Issued for teaching	A. Soottitantawat	—	—

Numbers are taken from the BFW Steam Heater Ver 1 simulator model; values marked assumed / TBC / HOLD are not design data.

CONTENTS

1 Purpose and scope 2

2 Design basis 2

3 Scenario screening (API 521) 3

4 Model relief runs 3

5 Sizing (API 520 Part I) 5

6 Relief device register 6

7 Vacuum breaker, thermal relief and vents 7

8 Inlet loss and back-pressure of PSV-100 7

9 Checks against the model and other documents 7

10 Engineering queries 7

Notes 8

1 Purpose and scope

This document is the relief load basis of the E-101 BFW steam heater: it lists every way the shell (flash steam, PSV-100) and the tubes (BFW) can be over- or under-pressured, decides which cases are credible, gives the relief load and relieving conditions of each, sizes the devices with the API 520 Part I relations and lists them in a register with their discharge routing.

The simulator model has no relief device. The pressures quoted from the model are the ones a relief device would see; they come from copies of the model started at the design snapshot with the trips disabled.

Other layers of protection (BPCS, alarms, SIS I-100) are covered by the Safety layers, LOPA and SRS of SIF tabs and are not repeated here.

2 Design basis

Item	Value	Source
Shell design pressure = PSV-100 set	350 kPa g	Design tab (assumed); LOPA tab uses 300 (EQ-30)
Relieving pressure (10 % overpressure)	385 kPa g	set × 1.10
Relieving temperature	151.1 °C (λ 2,087 kJ/kg)	Tsat at the relieving pressure
Tube-side design pressure	1,100 kPa g	Design tab (assumed)
BFW maximum (pump shut-off)	1,030 kPa g	model K.Ptk + K.H0
Two-thirds rule	⅔ × 1,100 = 733.7 kPa g	shell design below → tube rupture credible
Steam header maximum	300 kPa g	assumed, top of the fault range (EQ-31)
Shell wetted area for fire	11.6 m²	Design tab geometry

3 Scenario screening (API 521)

#	Scenario	How it arises	Credible?	Relief load	Relieving conditions	Model (no PSV)	Governs?
S1	**Blocked condensate outlet**	V-101 or V-102 closed, Y-101 / ST-101 fully plugged; also the return header at its maximum back-pressure (160 kPa g).	Yes as a cause; pressure-limited	None: the shell can only rise to the header pressure, below the set pressure	—	198.9 kPa g, hotwell 81 %; back-pressure 160 → 181.8	No
S2a	**Control-valve failure: FV-104 fails open, header at design**	FV-104 wide open (positioner or signal failure); header 200 kPa g.	Yes as a cause; pressure-limited	None (shell below set)	—	166.8 kPa g, steam 13,432 kg/h	No
S2b	**FV-104 fails open at the assumed header maximum**	Header at 300 kPa g (assumed maximum, the top of the fault range) with FV-104 open.	Yes; below set	None: header 300 < shell design 350 kPa g	—	267.0 kPa g (crosses PSHH-100 250 at 6.25 s)	No
S2c	**FV-104 fails open with a header let-down failure**	Upstream let-down to the header fails; header toward 400 kPa g, beyond the fault range of the simulator.	Only if the header is not protected: HOLD (EQ-31)	10,552 kg/h steam (FV-104 wide open, shell at 385 kPa g)	385 kPa g / 151.1 °C	367.0 kPa g, above set at 8 s	No (would set the PSV-100 size: orifice Q)
S3	**Loss of cooling / condensing**	The shell is the heating side: losing the BFW stops condensation, but no heat source can push the shell above the header pressure.	Not applicable as a relief case	None (pressure-limited by the header); the hot-BFW hazard is TSHH-108 in the SIS	—	BFW lost, steam held: shell 200.0 kPa g, outlet 126 °C	No
S4	**External fire**	Pool fire under the exchanger; wetted area 11.6 m², F = 1 (no credit for insulation).	Yes	$Q = 43.2\ F\ A^{0.82} = 322.6$ kW → 557 kg/h steam	385 kPa g / 151.1 °C, λ 2,087 kJ/kg (10 % used; fire allows 21 %)	not modelled	No (sets PSV-100: orifice G)
S5	**Tube rupture**	One tube breaks; BFW at up to 1,030 kPa g (pump shut-off) enters the shell through both ends.	Yes: shell design 350 < ⅔ × 1,100 = 733.7 kPa g	22,452 kg/h BFW liquid (ΔP 645 kPa, C_d 0.7)	385 kPa g, liquid 18 °C	tube leak 1.5 kg/s: 118.7 kPa g (a leak, not a rupture)	Yes
S6	**Thermal expansion of blocked-in BFW**	Tube side isolated full of water while the shell is hot or steam is on (HAZOP R05).	Yes	≤ 3.08 m³/h (design duty as the upper bound of heat input)	TRV set 1,100 kPa g + 10 %	steam held, no flow: outlet 126 °C	No (own device: TRV, orifice D)
S7	**Vacuum on cooldown**	Steam cut (trip or stop) while the BFW keeps condensing the shell steam.	Yes, on every trip	air in-breathing; condensing 2.12 kg/s at 0 kPa g	VB-101 opens -2.5 kPa g (model)	VB-101 is modelled	No (own device: VB-101)
S8	**Utility failure**	Instrument air: FV-104 FC and XV-100 FC close the steam, FV-101A FO keeps the BFW. Power: pumps stop, FSLL-101 closes XV-100.	Yes as a cause	None: every fail position moves away from overpressure	—	as S3: 200.0 kPa g at most	No

Pressure-limited: a cause whose source pressure is below the set pressure gives no relief load. Only fire, tube rupture and a header let-down failure beyond the assumed maximum push the shell past the set pressure.

4 Model relief runs

Run (trips disabled, no PSV)	Peak shell kPa g	at s	> 250 at s	> 350 at s	Peak steam kg/h	Max level %	Peak BFW out °C
FV-104 fails open, header at design	166.8	73	—	—	13,432	5	74.3
FV-104 fails open, header high (300 kPa g)	267.0	77	6.2	—	20,047	5	79.9
FV-104 fails open, header let-down failure (toward 400 kPa g, beyond the fault range)	367.0	79	3.0	8.0	27,423	5	84.3
Condensate outlet blocked (strainer/trap fully plugged)	198.9	1,800	—	—	9,935	81	70.0
Condensate return back-pressure at maximum (160 kPa g)	181.8	1,800	—	—	9,935	40	70.0

Run (trips disabled, no PSV)	Peak shell kPa g	at s	> 250 at s	> 350 at s	Peak steam kg/h	Max level %	Peak BFW out °C
Tube leak into the shell (1.5 kg/s)	118.7	1,800	—	—	10,841	14	70.0
BFW flow lost with steam held (pump trips, spare unavailable)	200.0	72	—	—	9,935	5	126.0

5 Sizing (API 520 Part I)

Steam (API 520 Part I, Napier form): A = 190.5 W / (P_1 K_d K_b K_c K_N K_{sh}), A in mm², W in kg/h, P_1 in kPa a. External fire: W = 557 kg/h, P_1 = 350 × 1.10 + 101.3 = 486.3 kPa a, K_d 0.975, K_b 1 (conventional, back-pressure ≤ 10 %), K_c 1 (no disc upstream), K_N 1 (P_1 ≤ 10 339 kPa a), K_{sh} 1 (saturated) → A = 223.6 mm² → API 526 orifice **G** (324.5 mm²), rated 808 kg/h.

Liquid (API 520 Part I): A = 11.78 Q / (K_d K_w K_c K_v) √(G/ΔP), Q in L/min, ΔP in kPa. Tube rupture: Q = 22,452 / 998 = 22.50 m³/h = 375.0 L/min, G = 0.999, ΔP = 385 − 0 = 385 kPa, K_d 0.65, K_w 1, K_c 1, Re = 350,141 → K_v = 1.000 → A = 346.1 mm² → orifice **H**.

One PSV for both? Orifice G passes 21,048 kg/h of water at the same conditions, less than 22,452 kg/h, and a spring valve opens too slowly for the pressure wave of a broken tube: rupture disc RD-100 (K_d 0.62 method: A ≥ 363 mm², bore ≥ 21 mm; final size by transient analysis) is recommended, HOLD (R11).

Steam case FV-104 wide open with the header at 400 kPa g (if made credible by EQ-31): 10,552 kg/h → A 4,239 mm², orifice Q.

API 526 letter	D	E	F	G	H	J	K	L	M	N	P	Q	R	T
Area mm²	71.0	126.5	198.1	324.5	506.5	830.3	1,185.8	1,840.6	2,322.6	2,800.0	4,116.1	7,129.0	10,322.6	16,774.2

6 Relief device register

Tag	Device	Protects	Set / opens	Governing case and load	Size	Discharge to	Status
PSV-100	Conventional spring PSV, steam	E-101 shell	350 kPa g (assumed = shell design; P&ID: TBC)	External fire 557 kg/h steam at 385 kPa g / 151.1 °C	A 223.6 mm² → G (324.5 mm²); inlet 1½ in, outlet 3 in (TBC)	Safe location clear of walkways, with silencer (R17)	Existing; set TBC (EQ-33); size HOLD on EQ-31
RD-100	Rupture disc, liquid service	E-101 shell against tube rupture	burst ≤ 350 kPa g (TBC)	Tube rupture 22,452 kg/h BFW at 18 °C	A ≥ 363 mm² (bore ≥ 21 mm), final by transient analysis	Closed drain / blowdown pit, not an open sewer (R16, R17)	Recommended, HOLD (R11, EQ-32)
TRV-101 (tag TBC)	Thermal relief valve	E-101 tubes when blocked in	1,100 kPa g (tube design)	Thermal expansion ≤ 3.08 m³/h	A 26.4 mm² → D	Drain or safe location	Recommended (R05)
VB-101	Vacuum breaker	E-101 shell against vacuum	opens -2.5 kPa g (model)	Condensing 2.12 kg/s after XV-100 closes (BFW still flowing)	bore ≈ 194 mm at 20 kPa (conservative basis), TBC	Air inlet with screen, away from the vent and PSV discharges	Existing; size and shell vacuum rating TBC (R14, EQ-34)
V-104 / 1"-VT-101	Manual air vent	Non-condensables at the far end of the shell	manual (crack ≤ 10 % in service)	Air at start-up until dry steam blows	1 in	Safe location with silencer (R17)	Existing
RO-101 (rev B)	Continuous vent orifice	Non-condensables in service	always open	72 kg/h = 0.73 % of the steam (model)	inside the 0.5–1 % band 50–99 kg/h	Via 1"-VT-101	Option (rev B)
AAV-101 (rev C)	Automatic air vent + TI-102 / TAL-102	Non-condensables in service	self-acting (opens when the vent line is cool)	Gas only; closes on steam	—	Via 1"-VT-101	Option (rev C)

7 Vacuum breaker, thermal relief and vents

VB-101: after XV-100 closes the BFW keeps flowing and condenses the shell steam. At 0 kPa g (Tsat 100.0 °C) the tubes condense 2.12 kg/s ($Q = F\ cp\ (Tsat - Tin)(1 - e^{-NTU})$ = 4,793 kW, NTU 0.70). Replacing that steam volume (3.61 m³/s) with air at a 20 kPa vacuum (Cd 0.6, assumed) needs A = 29,524 mm², bore ≈ 194 mm. This conservative basis shows why a full-vacuum shell is often chosen instead (HAZOP R14, EQ-34). VB-101 opens at −2.5 kPa g in the model.

Tube-side TRV: $q = \beta\ \Phi\ /\ (\rho\ cp)$ = 0.00058 × 6,040 kW / (978 × 4.191) = 3.08 m³/h with the design duty as the upper bound; at 1,100 kPa g + 10 %: A = 26.4 mm² → orifice D (HAZOP R05).

Vents: V-104 on 1"-VT-101 at the far end of the shell; rev B RO-101 passes 72 kg/h = 0.73 % of the steam (model), inside the 0.5–1 % band 50–99 kg/h; rev C AAV-101 with TI-102 / TAL-102. All discharges to a safe location with a silencer (HAZOP R17).

8 Inlet loss and back-pressure of PSV-100

At the rated capacity of orifice G (808 kg/h). Inlet: 1½ in sch 40 (ID 40.9 mm), 0.5 m, K 0.5, steam 2.48 kg/m³: v 68.7 m/s, ΔP 4.42 kPa = 1.3 % of set (limit 3 %). Outlet: 3 in sch 40 (ID 77.9 mm), 6 m, K 1.5, steam 0.538 kg/m³ (≈ 135 °C after throttling, assumed): v 87.5 m/s, built-up back-pressure 6.02 kPa = 1.7 % of set (limit 10 % for a conventional valve).

Incompressible estimate (Darcy, Haaland friction factor, roughness 0.045 mm) with the density at each end; adequate when the result is a few per cent.

9 Checks against the model and other documents

Check	This document	Model / other document	Agrees
PSV-100 set ≤ shell design pressure	set 350 kPa g (assumed, TBC)	Design tab shell design 350 kPa g	✓
Shell design pressure, Design tab vs LOPA / Safety layers tabs	350 kPa g (Design tab basis)	300 kPa g (LOPA tab and its deck)	✗ EQ-30 raised
Relieving pressure = set + 10 % overpressure	385 kPa g	Design tab 385 kPa g	✓
Relieving temperature = T_{sat}(relieving pressure)	151.1 °C	Design tab 151.1 °C	✓
Governing case equals the Design tab	tube rupture	Design tab: tube rupture	✓
Tube-rupture load	22452 kg/h liquid	Design tab 22452 kg/h	✓
External-fire load	557 kg/h steam	Design tab 557 kg/h	✓
Two-thirds rule: tube rupture credible	350 < 733.7 kPa g → credible	Design tab: credible	✓
Tube design ≥ BFW maximum (pump shut-off)	1100 kPa g	K.Ptk + K.H0 = 1030 kPa g	✓
Header maximum below shell design (steam overpressure not credible)	300 kPa g (assumed)	shell design 350; Design tab: not credible	✓
Model: FV-104 fails open, header at design, stays below set	peak 166.8 kPa g	set 350	✓
Model: FV-104 fails open at the assumed header maximum stays below set	peak 267.0 kPa g	set 350	✓
Model: header let-down failure (toward 400, beyond the fault range) stays below set	peak 367.0 kPa g, crosses 350 at 8 s	set 350	✗ EQ-31 raised
Model: blocked condensate outlet and lost BFW are pressure-limited	198.9 / 200.0 kPa g	header 200 < set 350	✓
PSHH-100 trips below the PSV set (≤ 90 % of set)	trip 250 kPa g	0.9 × 350 = 315 kPa g	✓
PI-100 high alarm below the PSHH-100 trip	alarm 200 kPa g	trip 250 kPa g	✓
One PSV (sized for the steam cases) also passes the tube-rupture liquid	PSV-100 G passes 21048 kg/h water	tube rupture 22452 kg/h, and a spring valve is slow for a liquid surge	✗ EQ-32 raised
PSV-100 set pressure stated on the P&ID	set 350 kPa g used here (assumed)	BFWH-V1-PID-001 rev D: "SET TBC"	✗ EQ-33 raised
PSV-100 inlet loss ≤ 3 % of set	4.4 kPa = 1.3 %	API 520 Part II guidance (3 %)	✓
PSV-100 built-up back-pressure ≤ 10 % of set (conventional valve)	6.0 kPa = 1.7 %	10 % for a conventional valve	✓
Continuous vent RO-101 inside 0.5–1 % of the steam	72 kg/h (model, rev B)	Design tab 50–99 kg/h	✓
Tube-side TRV set above the pump shut-off (no lift in service)	set 1100 kPa g (= tube design)	shut-off 1030 kPa g	✓

10 Engineering queries

EQ	Subject	Query	Status
EQ-30	E-101 shell design pressure	The Design tab sizes PSV-100 with a shell design pressure of 350 kPa g; the LOPA and Safety layers tabs (and the LOPA deck) use 300 kPa g. Confirm one shell design pressure (= PSV-100 set) and align all documents. At 300, the FV-104-open case at the 300 kPa g header maximum (model 267 kPa g) still stays below set, but the margin to PSHH-100 (250) shrinks.	Open
EQ-31	Steam header maximum pressure	The relief basis rests on a header maximum of 300 kPa g (assumed, top of the simulator fault range). With a let-down failure toward 400 kPa g the model shell reaches 367 kPa g and crosses 350 at 8 s; FV-104 wide open would then need 10 552 kg/h of relief (orifice Q). Confirm the header maximum and its protection (pressure control or relief on the header, HAZOP R08/R09) or size PSV-100 for this case.	Open
EQ-32	Tube-rupture relief device	Tube rupture governs (22 452 kg/h BFW, two-thirds rule 733.7 > 350 kPa g). PSV-100 sized for steam (orifice G) passes only about 21 050 kg/h of water and is too slow for the pressure wave of a broken tube. Rupture disc RD-100 recommended (HAZOP R11): confirm by transient analysis, set burst pressure and discharge route. HOLD.	Open
EQ-33	PSV-100 set pressure	BFWH-V1-PID-001 rev D shows PSV-100 'SET TBC'. This document uses 350 kPa g (= shell design, Design tab). Confirm the set pressure from the E-101 datasheet and update the P&ID and the C&E.	Open
EQ-34	VB-101 size and shell vacuum rating	Replacing the full condensing rate after XV-100 closes (2.12 kg/s at 0 kPa g) needs a breaker bore of about 194 mm at 20 kPa vacuum. Confirm the shell vacuum rating (-30 kPa g assumed on the Safety layers tab) or full-vacuum design, and the VB-101 size and test interval (HAZOP R14).	Open

Notes

- Numbers are computed by tools/mk_data.js from the simulator model data (unit.json, runs.json) with the same functions as the Relief & vent tab.
- Assumed values: shell design 350 kPa g, tube design 1 100 kPa g, header maximum 300 kPa g, overpressure 10 %, Cd 0.7 at the tube break, fire factor F = 1, Kd 0.975 (steam) / 0.65 (liquid) / 0.62 (disc), β 0.00058 1/K, VB-101 ΔP 20 kPa and Cd 0.6, line lengths and fittings of section 8.
- TBC / HOLD: PSV-100 set (EQ-33), header maximum (EQ-31), RD-100 (EQ-32), shell vacuum rating and VB-101 size (EQ-34), shell design pressure (EQ-30).